MIC-501T UPDATED

SUBJECTIVES FINALTERM- 100%

MICROBIOLOGY

BY SULMAN ALI GUJJAR

0341-7547044

CURRENT SUBJECTIVES

Antigen and Epitope.

Antigen:

A foreign substance that provokes immunity is called an antigen. The definition of antigen is a harmful substance which enters the body which causes the body to make antibodies as a response to fight off disease. More specifically called as an immunogen. Proteins are best antigen, though glycoproteins and lipoproteins can also act as antigens. Carbohydrates and lipid alone are poor antigens.

Epitopes:

Specific regions on an antigen. In other words, any antigen has multiple regions each one of those acts as a small antigen. These are called epitopes. These epitopes are also called as antigenic determinants. o Most antigens have MW >10,000 D. Molecules that are less than 10,000 D are generally poor antigen.

Role of Thin Clonal Selection?

According to this theory, B cells are generated in the bone marrow randomly without seeing the antigen. When such a B cell encounters its specific antigen in the body and start secreting antibodies. A beautiful example to understand clonal selection theory is like going to a shoe store and select a shoe size that fits you very well. Note, that the shoe that you selected was made randomly without any regard to your foot size.

Describe 5 bacteria which need oxygen for growth?

1= Nocardia asteroids, 2= Mycobacterium tuberculosis, 3= Bacillus anthracis , 3= Pseudomonas, 4= aeruginosa, 5= Neisseria gonorrhea.

Define and Types of Algae. / describe Rhodophyta, bacillariophyta, chlorophyta, phaeophyta.

Algae are simple eukaryotic organisms lacking roots, stems, or leaves, commonly found in water. They exhibit a wide diversity of forms, ranging from single-celled to multicellular structures. Algae are classified into several types based on characteristics such as color and cell composition:

1. **Phaeophyta (Brown Algae):** Multicellular, brownish, containing cellulose and alginic acid cell walls, and capable of rapid growth.
2. **Rhodophyta (Red Algae):** Predominantly multicellular, red in color, with cellulose cell walls, storing glucose polymer for agar production.

3. **Chlorophyta (Green Algae):** Unicellular or multicellular, containing chlorophyll a and b, believed to be ancestral to terrestrial plants.
4. **Bacillariophyta (Diatoms):** Unicellular, brownish, with pectin and silica cell walls, storing oil and producing domoic acid.
5. **Dinoflagellates:** Unicellular plankton with cellulose in the membrane, possessing chlorophyll a and c, and carotene, storing starch.

Inflammation and Write Its Consequence? / Characteristics:

Inflammation is a local defensive response to tissue damage, manifesting as redness, pain, swelling, and heat. It involves the release of inflammatory mediators like histamine, kinins, prostaglandins, leukotrienes, and cytokines, including TNF-alpha. These mediators induce the release of acute phase proteins from the liver. The consequences of inflammation include mobilizing immune cells to the site, initiating tissue repair, and isolating harmful agents. While essential for healing, chronic inflammation can lead to various diseases.

Types of Symbiosis?

Interaction between two different organisms living in close physical association is called **symbiosis.**

1. **Mutualism:** Both are benefited Parasitism: one is harmed.
2. **commensalism:** One is benefited, other is unharmed.
3. **Parasitism:** one is harmed.

Characters of antigen?

Antigens are molecules recognized by the immune system, often found on pathogens. They possess specific molecular features called epitopes that induce immune responses. Antigens can be proteins, carbohydrates, nucleic acids, or lipids, provoking the production of antibodies and cellular immune responses.

Define Culture, Culture Medium, Inoculation?

Microbial growth in the lab is called a culture. A nutrient material that supports the growth of microbes in the lab is called a culture medium. Microbes introduced into a culture medium that initiate growth of organisms are known as inoculum.

Define Five Kingdom Classification of Organisms.

The Five Kingdom Classification classifies organisms into five groups: Monera (prokaryotic bacteria), Protista (single-celled eukaryotes), Fungi (multicellular, absorptive heterotrophs), Plantae (multicellular, photosynthetic autotrophs), and Animalia (multicellular, ingestive heterotrophs). This system, proposed by R.H. Whittaker, categorizes living organisms based on their cellular structure, nutrition, and reproduction methods.

How disease effect normal to healthy person:

Disease can adversely impact a normal, healthy person by disrupting physiological functions. Symptoms, ranging from mild to severe, can arise, affecting daily activities and overall well-being. The immune system responds to combat the pathogen, potentially causing fatigue, fever, and inflammation. If left untreated, diseases can lead to complications, organ damage, or chronic conditions, impacting the individual's physical and mental health. Effective medical intervention is crucial to restore health and prevent long-term consequences in otherwise healthy individuals.

Viral Replication and Its Mechanism? Stages of Viral Replication?

Viral replication involves six steps:
1= Attachment, 2= Penetration, 3= Uncoating, 4= Replication, 5= Assembly, 6= Release.
During uncoating, replication, and assembly, the viral DNA or RNA incorporates itself into the host cell's genetic material and induces it to replicate the viral genome.

Write three characteristic of bacterial cell wall?

1. **Peptidoglycan Structure:** Bacterial cell walls contain peptidoglycan, a polymer with alternating sugar chains, providing rigidity.
2. **Protection:** The cell wall protects against osmotic stress.
3. **Shape Determination:** It contributes to bacterial shape and maintains structural integrity.

Diff between Northern , Southern & Western blot:

1= Southern blotting: Southern blot is a method used in molecular biology for detection of a specific DNA sequence in DNA samples.

2= Northern blotting: The northern blot is a technique used in molecular biology research to study gene expression by detection of RNA (or isolated mRNA) in a sample.

3= Western blotting: The western blot (sometimes called the protein immunoblotting) is a widely used analytical technique used to detect specific proteins in a sample of tissue homogenate or extract.

Drugs of cell wall inhibition:

Drugs that inhibit bacterial cell wall synthesis include antibiotics like penicillins, cephalosporins, and vancomycin. These drugs interfere with the formation of the bacterial cell wall, disrupting its structural integrity and leading to the lysis of bacterial cells.

How bacteria defend the antimicrobial drugs:

Bacteria can defend against antimicrobial drugs through mechanisms such as developing resistance genes, modifying drug targets, reducing drug entry, or producing enzymes that inactivate the drugs. These adaptive strategies contribute to the emergence of antibiotic-resistant bacterial strains.

Describe Two Transported System in The Blood and Why It Is Necessary?

Cardiovascular and lymphatics are the two transport systems of the body by which cells of the immune system patrol the whole body.

1= Cardiovascular system consists of the heart and associated blood vessels (arteries and veins) in which blood circulates.

2= Lymphatics are channels that start from tissue spaces and collect tissue fluid (called lymph) from tissue spaces and bring them back to the heart. Along these lymphatic channels, there are strategically placed lymph nodes which filter lymph for microbes much like blood is filtered in the Spleen.

Significant and characteristics of media:

Bacterial media, vital for cultivating bacteria, vary in composition and purpose. Selective media encourage specific bacterial growth, while differential media distinguish between bacterial species based on metabolic traits. Enriched media contain additional nutrients for fastidious organisms. Proper selection aids in accurate bacterial identification and cultivation.

Differentiate dark Field, bright filed, fluoricent filed.

1. Dark Field Microscopy: Illumination from the sides makes specimens appear bright against a dark background, enhancing contrast for live, unstained samples.

2. Bright Field Microscopy: Illumination from below renders specimens visible against a bright background, suitable for stained specimens with high contrast.

3. Fluorescence Microscopy: Utilizes fluorescence from labeled molecules, enhancing contrast and allowing specific visualization of structures in cells or tissues.

Grow and preserve L-Broth:

L-Broth (Lysogeny Broth): L-Broth is a nutrient-rich liquid medium used to culture bacteria. To prepare, mix tryptone, yeast extract, and sodium chloride in water, adjust pH, and sterilize.

For solid L-Broth (Lagar), add agar before sterilization. To grow bacteria, inoculate L-Broth with a small culture, incubate, and monitor growth. To preserve, store cultures in L-Broth with glycerol at -80°C. This nutrient-rich medium supports bacterial growth, facilitating research, and long-term storage preserves bacterial strains for future use.

Define vaccine:

A vaccine is a biological preparation that stimulates the immune system to recognize and defend against specific pathogens, typically viruses or bacteria. It contains weakened, killed, or parts of the pathogen, training the immune system to respond effectively, providing immunity upon subsequent exposure.

Who name the small pox and when?

Lady Mary Wortley Montagu named the smallpox in 1717, while Edward Jenner developed the smallpox vaccine in 1796.

Definition Karyogamy, Plasmogamy and meiosis/ Sexual reproduction in fungi:

Karyogamy: Fusion of two haploid nuclei to form a diploid nucleus, completing the sexual reproduction cycle.

Plasmogamy: Fusion of cytoplasm from two mating cells without nuclear fusion, often preceding karyogamy in fungal sexual reproduction.

Meiosis: Cell division process reducing chromosome number by half, generating haploid cells, crucial for sexual reproduction and genetic diversity.

Describe disk diffusion theory:

Disk diffusion theory is a method to test antimicrobial susceptibility. Paper disks with known antimicrobial agents are placed on agar plates inoculated with bacteria. Inhibition zones around the disks indicate the effectiveness of the antimicrobial agents against the bacterial strain.

In this:

Filter paper disks soaked with anti-bacterial

Plates are incubated at 37oC and results are recorded.

Types of microscope:

Give Five Types of Microscopy:

1= Darkfield Microscopy: Enhances contrast by illuminating specimens against a dark background., 2= Phase Contrast Microscopy: Accentuates phase shifts in light passing through transparent specimens., 3= Fluorescent Microscopy: Uses fluorescent dyes to visualize specific cellular structures., 4= Transmission Electron Microscopy: Achieves high resolution by transmitting electrons through thin specimens., 5= Scanning Electron Microscopy: Provides detailed surface images using electrons.

Consequences of viral infections:

Consequences of viral infections encompass a spectrum of effects on host cells. These include the lytic phase, causing cell lysis and degenerative changes, both microscopic and macroscopic (cytopathic effects). Virulent infections result in cell lysis, while latent infections involve suppressed viral replication. Persistent infections exhibit slow viral replication. Additionally, viral infections can lead to cellular transformation, contributing to the development of tumor cells, highlighting the diverse and complex outcomes associated with viral interactions within host organisms.

Define Magnetosomes:

Magnetosomes are inclusions of iron oxide surrounded by invaginations of the plasma membrane. Typically found in Gram-negative bacteria, they function as magnets, enabling bacteria to adhere to iron-containing rocks for nutrition. Additionally, magnetosomes assist in decomposing hydrogen peroxide (H2O2), a toxic substance for cells.

Antibiotic for inhibition of bacterial cell wall protein and Nucleic acid:

Bacterial Cell Wall Inhibition: Penicillins, such as amoxicillin, inhibit bacterial cell wall synthesis by targeting enzymes involved in peptidoglycan formation, leading to cell wall disruption and bacterial lysis.

Nucleic Acid Inhibition: Ciprofloxacin, a fluoroquinolone antibiotic, inhibits bacterial DNA gyrase and topoisomerase IV, preventing DNA replication and transcription, ultimately hindering bacterial growth.

Basic feature of fungi:

Fungi are eukaryotic organisms with cell walls made of chitin. They reproduce via spores and exhibit diverse forms, including molds and yeasts. Fungi are vital decomposers, forming mutualistic or parasitic relationships. They play roles in nutrient cycling, medicine, and food production.

Why virus need live organism. which type of cell is used by virus:

Fungi are eukaryotic organisms with cell walls made of chitin. They reproduce via spores and exhibit diverse forms, including molds and yeasts. Fungi are vital decomposers, forming mutualistic or parasitic relationships. They play roles in nutrient cycling, medicine, and food production. An example is the influenza virus, which infects respiratory epithelial cells in the respiratory tract of animals.

Name Three Nucleic acid-based method of bacterial identification:

Nucleic acid base microbial detection methods:

1= DNA probes, 2= DNA chips, 3= PCR.

Flagellum and its function:

A flagellum is a whip-like appendage found in some cells, such as bacteria and sperm. Its primary function is to propel the cell through its environment, aiding in movement and facilitating the cell's response to external stimuli.

Define Algae and its structure.

The definition of algae is a single or multi-cellular organism that has no roots, stems or leaves and is often found in water. These are simple eukaryotic cells.

Structure: Body of multicellular alga such as seaweed is called a thallus which consists of branched holdfasts (anchor alga to rock) stem like hollow stipes and leaf like blades. There is no vascular tissue in these algae. Surrounding water provides the support for the thallus. Some algae have a gas filled body inside them which keeps them floating in the water. This gas filled structure is called pneumatocyst or float.

Types of microbes based on oxygen requirements?

1. **Obligate Aerobes:** These organisms exclusively require oxygen for growth and cannot thrive in its absence. In a nutrient agar tube, their growth is confined to the surface, exemplified by Mycobacterium tuberculosis.
2. **Facultative Anaerobes:** Preferring aerobic respiration, these organisms can switch to anaerobic modes in the absence of oxygen, distributing growth throughout the tube with enhanced development near the surface when oxygen is available.
3. **Obligate Anaerobes:** Unable to use oxygen, these organisms also cannot grow in its presence due to the toxic compounds oxygen forms. Strict anaerobes lack mechanisms to eliminate these toxic compounds.
4. **Aerotolerant Aerobes:** These organisms neither use oxygen nor are affected by its presence, possessing efficient systems to manage and eliminate toxic oxygen compounds.

List four phases of growth curve.

In a batch culture, organisms go through various phases of growth. If we follow this growth with respect to time, and make a graph of this growth pattern, it will give us what we call as a growth curve. There are four distinct phases of this curve.

1= Lag phase, 2= Log phase, 3= stationary phase, 4= Death phase.

How microbes harm the organisms:

Microbes can harm organisms through infections and diseases. Pathogenic microbes, like bacteria, viruses, and fungi, invade host organisms, disrupt normal physiological functions, and induce inflammation. They may damage tissues, interfere with immune responses, and cause various health complications.

what is the purpose and use of western blotting.

Western blotting is a technique to detect and analyze specific proteins in a sample. It involves transferring proteins from a gel to a membrane, followed by antibody detection. It is used for protein identification, quantification, and studying protein expression in biological samples.

Hyphae, mycelium and thallus.

Hyphae: Thread-like structures in fungi, forming the basic structural unit for growth and nutrient absorption.

Mycelium: A network of interconnected hyphae, constituting the vegetative body of fungi, responsible for nutrient uptake and environmental interaction.

Thallus: The undifferentiated body of lower plants, algae, or fungi, lacking specialized structures.

Define Metachromatic granules:

Metachromatic granules, also known as volutin, are large inclusions within bacterial cells. They stain red with methylene blue due to their unique property of displaying color changes during staining. Composed of inorganic phosphates, these granules are characteristic features observed in bacteria, such as Corynebacterium diphtheriae.

Three types of flagella:

1. **Bacterial Flagella:** Helical filamentous structures, common in bacteria, propel the cell.
2. **Archaeal Flagella:** Distinct from bacterial flagella, composed of multiple subunits, enabling movement in archaea.
3. **Eukaryotic Flagella:** Found in certain eukaryotic cells, like sperm, with a whip-like motion for cell motility.

Name the bacteria that inhibit cell wall synthesis:

Bacteria that inhibit cell wall synthesis include:

1. **Penicillin-producing bacteria:** Penicillin interferes with peptidoglycan formation.
2. **Cephalosporin-producing bacteria:** Cephalosporins disrupt bacterial cell wall synthesis.
3. **Vancomycin-producing bacteria:** Vancomycin interferes with peptidoglycan synthesis. These antibiotics are effective against bacterial infections.

Mechanism of defensive metabolism:

Defensive metabolism is a plant's response to stress or herbivore attacks. It involves the production of secondary metabolites, like phenols and terpenoids, deterring herbivores or pathogens. Plants activate specific genes, redirecting metabolic pathways to synthesize these compounds. Defensive metabolites can act as antifeedants or toxins, contributing to the plant's defense against potential threats.

Acid Fast Stanning:

Some bacteria such as Mycobacteria have a waxy material in their cell wall. Specific name for this waxy substance is mycolic acid.

Principle of Acid-Fast Staining:

Mycobacteria are lipophilic, not easy to stain but once stained, they are resistant to acidic alcohol decolorization process.

Procedure for Acid-Fast Staining:

- ❖ Make a smear
- ❖ Carbolfuchsin on slides
- ❖ Heat the slide for 5-10 min
- ❖ Wash with water and then with acid alcohol for 1 min.
- ❖ Counterstain with methylene blue for 1 min
- ❖ See under the microscope.

Acid fast bacteria have mycolic acid that can be stained by heating the organisms with the stain.

Enlist steps of Gram Staining:

1. Primary Stain (Crystal Violet): Colors all cells.
2. Mordant (Iodine): Enhances crystal violet retention.
3. Decolorization (Alcohol or Acetone): Removes stain from some cells.
4. Counterstain (Safranin): Colors decolorized cells.
5. Wash and Dry: Final steps to observe Gram-positive (purple) and Gram-negative (pink) bacteria.

PCR denaturing, annealing and extended stage note.

PCR goes through 3 step cycles for about 30 to 35 times. Each cycle consists of:

Denaturing stage (about 95oC): when the double-stranded template DNA is heated to separate it into two single strands.

Annealing (50 to 56oC): in which step temperature is lower and DNA primers attach to DNA template

Extending stage (72oC): when the temperature is raised and the new strand of DNA is made by the Taq polymerase enzyme.

Define Spontaneous Generation:

(Aristotle) Spontaneous generation was a once-held belief that living organisms could arise spontaneously from non-living matter. This concept, prevalent until the 19th century, proposed that life could emerge without pre-existing life. However, advancements in microbiology, particularly Pasteur's experiments, disproved this notion, establishing the principle of biogenesis.

What is probe uses of probe In laboratory:

A probe is a labeled DNA or RNA molecule used in molecular biology to detect specific nucleotide sequences. In laboratories, probes are crucial for DNA/RNA hybridization studies, identifying genes, diagnosing diseases, and studying genetic material by binding to complementary sequences.

Five category of bacteria which growth oxygen requirements?

1. **Aerobes**
2. **Anaerobes**
3. **Facultative Anaerobes**
4. **Microaerophiles**
5. **Aerotolerant Anaerobes**

Pathogens microorganisms can cause damage the body:

Pathogenic microorganisms, including bacteria, viruses, fungi, and parasites, can cause harm by infecting and multiplying within the body. They produce toxins, induce inflammation, and disrupt normal physiological functions, leading to diseases. The immune system responds, but pathogens may evade or overwhelm defenses, causing illness and potential organ damage.

what the Routes of bacteria or infection or disease to enter the body:

1. **Respiratory Tract:** Inhaling airborne bacteria.
2. **Gastrointestinal Tract:** Ingesting contaminated food or water.
3. **Genitourinary Tract:** Through sexual contact or contaminated fluids.
4. **Breaks in the Skin:** Wounds or injuries allowing bacterial entry.
5. **Vertical Transmission:** From mother to child during childbirth.

Name antibiotics which inhibit bacterial cell wall synthesis, dna synthesis, protein synthesis:

1. **Cell Wall Synthesis Inhibitors:** Penicillin, Vancomycin.
2. **DNA Synthesis Inhibitors:** Ciprofloxacin, Metronidazole.
3. **Protein Synthesis Inhibitors:** Tetracycline, Erythromycin. These antibiotics target specific bacterial processes, disrupting cell wall construction, DNA replication, or protein synthesis, leading to bacterial growth inhibition or death.

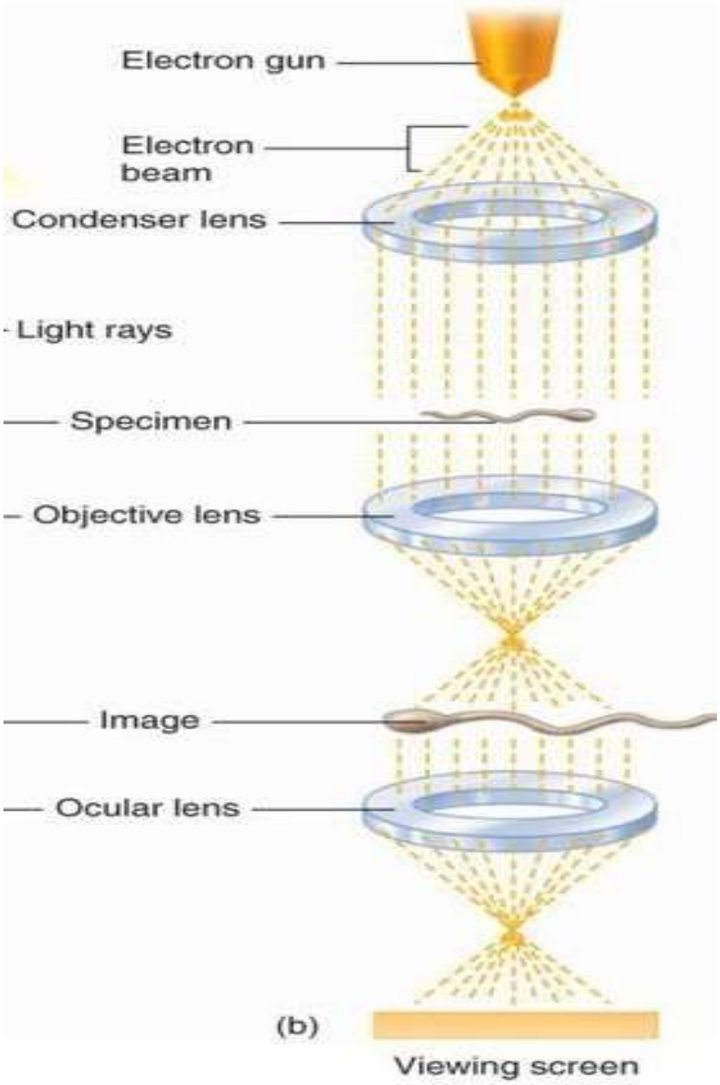
2 transport systems and why are they necessary

1. **Active Transport:** Utilizes energy to move molecules against their concentration gradient. Essential for nutrient uptake, enabling cells to acquire nutrients even when scarce, and expel waste products.

2. **Passive Transport (Diffusion):** Movement of molecules along their concentration gradient, requiring no energy input. Facilitates the exchange of ions and gases, ensuring equilibrium within cells.

3. Both systems are vital for microbial survival, maintaining homeostasis and supporting cellular functions.

4. Label Electron Transmission Microscope:



Define Morbidity,or mortality and pathogenicity?

1. Morbidity: Incidence of a disease
2. Morbidity rate: Number of people affected in relation to the total population in a given time period
3. Mortality: Deaths from a disease
4. Mortality rate: Number of deaths from a disease in relation to the population in a given time
5. Pathogenicity: The ability to cause disease
6. Virulence: The extent of pathogenicity

Define any Three shapes of bacteria?

1= Coccus These are circular or spherical in shape. 2= Bacillus These rod-shaped bacteria. 3=

Spiral These are curved shaped bacteria. They are further divided into 3 more subgroups. Vibrio: curved rods Spirillum: Helical but rigid Spirochete: Helical but flexible Spirochetes move by axial filaments which are enclosed by the outer membrane.

Viruses, Virion, And Virusoid And Prion?

1. Viruses: Acellular organisms — Proteins and nucleic acids
2. Viroids: Composed only of circular ssRNA (potato spindle tuber viroid)
3. Virusoids: Contain circular ssRNA, need helper viruses for replication and encapsidation; also called satellite viruses
4. Prions: Infectious protems

Define epidemic, pandemic, endemic.

Endemic, Epidemic and pandemic.

1. **Endemic:** When a disease is consistently present, but with low incidence, we call it an endemic disease. Common cold is a typical example.
2. **Epidemic:** Within an area, unusually high incidence of a disease is called an epidemic. Influenza is a good example.
3. **Pandemic:** When a disease assumes a global occurrence or becomes worldwide, it is called an epidemic. AIDS is a good example of such as epidemic.

Three differences between eukaryotes and prokaryotes.

Three differences between prokaryotes and eukaryotes: 1= Prominent nucleus is found in eukaryotes while it is absent in prokaryotes. 2= Membrane bound organelles are found in eukaryotes and absent in prokaryotes. 3= No histones are found in prokaryotes while present in eukaryotes. 4= Examples of prokaryotes include bacteria and archaea, while all organisms in the domain Eukarya, such as plants and animals, are eukaryotes.

5 characteristics of bacterial cell membrane.

The cell membrane serves as a selective permeable barrier, facilitating passive and active transport. In microbes, it orchestrates respiration and photosynthesis. Bactoprenol aids in transporting cell wall components, while the plasma membrane's receptors initiate gene expression signals through ligand binding, regulating various cellular functions.

Steps in glucose respiration

Respiration of glucose takes place in three general steps:

Glycolysis: Oxidation of glucose to pyruvic acid

Krebs cycle: Oxidation of acetyl CoA to CO2

Electron Transport Chain: Coenzymes that carry electrons from Krebs cycle or glycolysis are oxidized to create ATP.

Three types of passive movement

1. **Simple Diffusion:** Movement of molecules from an area of higher concentration to lower concentration across a semi-permeable membrane.
2. **Facilitated Diffusion:** Molecules move across a membrane through protein channels.
3. **Osmosis:** Movement of water across a selectively permeable membrane.

PAMPs:

Pathogen associated molecular patterns (PAMPs): Formed by LPS, flagellin, peptidoglycan, DNA, RNA.

Parts of microscope

1. Eyepiece (ocular): The lens closest to the eye.
2. Objective lenses: Magnifying lenses closest to the specimen.
3. Nosepiece (turret or revolver): Rotating device holding multiple objective lenses.
4. Stage: Platform holding the specimen.
5. Condenser: Focuses light onto the specimen.
6. Illuminator: Light source.
7. Diaphragm: Controls the amount of light.
8. Coarse and fine focus knobs: Adjust focus.

9. Arm: Supports the upper parts.
10. Base: Provides stability.

Definition of fimbriae it's location and function?

Hair-like small appendages on G-neg cells, Composed of pilin, a protein Can be at one pole or around the entire cell , Used for attachment to surfaces or epithelial cells

Fimbria: Singular

Example Neisseria gonorrhoeae

Economic importance of fungi?

Fungi hold significant economic importance. They contribute to food production (yeast in baking), pharmaceuticals (antibiotics like penicillin), industrial processes (fermentation in brewing), and agriculture (mycorrhizal fungi aiding plant growth). Additionally, they are used in research and biotechnological applications, showcasing their diverse economic contributions.

factors determine the composition and distribution of microbiota

Availability of various nutrients determines the type of organisms present in a system. Similarly, physical and chemical factors also influence the type of microbiota. Other factors include pH, O2, and CO2.

Nutrients

Physical and chemical factors • pH, O2, CO2

Host defenses • Stress, age

Conidia, Arthroconidia, bladtoconidia?

1. **Conidia:** Chain of spores at the end of a conidiophore Aspergillus
2. **Arthroconidia:** spores formed by fragmentation of septate hypha
3. **Blastoconidia:** spores as budding of the parent cells
4. **Chlamydioconidia:** spores formed by rounding and enlargement within a hyphal segment

How bacteria get sulfur for their growth?

Bacteria obtain sulfur for growth through various mechanisms. Some bacteria can assimilate sulfur from inorganic compounds like sulfates or hydrogen sulfide, while others may engage in sulfur-containing amino acid uptake from organic sources in their environment.

procedure of western blotting?

In Western blotting, proteins are separated by gel electrophoresis, transferred to a membrane, and blocked. The membrane is then probed with a specific antibody, followed by secondary antibody binding to the primary antibody. Chemiluminescent or fluorescent signals indicate protein presence. Image capturing and analysis reveal protein bands, aiding in protein identification, quantification, and characterization.

Define following term :

1= Gene: The basic unit of heredity passed from parent to child. 2= Chromosome: threadlike structures made up of protein and DNA that carry genomic information from cell to cell. 3= Genotype: Genotype is the genetic makeup of an individual cell or organism that determines or contributes to its phenotype, 4= Phenotype: all observable characters that influenced by genotypes and environment.

PAST SUBJERCTIVE

Morbidity, Mortality, and Pathogenicity.

1. **Morbidity:** Incidence of a disease
2. **Mortality:** Deaths from a disease
3. **Pathogenicity:** The ability to cause disease.

How Microorganism Cause Damaged the Body?

- **Nutrient depletion:** Siderophores for iron ▪ Lactoferrin ▪ Transferrin ▪ ferritin
- **Direct damage**: Bacteria that grows in host cells rupture the cells and spread further.
- **Toxin production:** • Exotoxins • Antitoxins • Toxoid • A-B toxins

Inhibits Growth Cell Wall Drug Name?

Penicillin's and cephalosporins are the major antibiotics that inhibit bacterial cell wall synthesis.

Antigen Presenting Cell.

These cells are able to present antigens to B cells for making antibodies by B cells. There are three cells that fall into this category: Dendritic cells, macrophages, and B cells.

Define Mold, Yeast, Mycology.

1. **Mold:** Filamentous fungi
2. **Yeast:** unicellular fungus
3. **Mycology:** Study of fungi

How Microbes Develop Resistance in Antimicrobial?

Antimicrobial resistance is accelerated when the presence of antibiotics and antifungals pressure bacteria and fungi to adapt. Antibiotics and antifungals kill some germs that cause infections, but they also kill helpful germs that protect our body from infection. The anti-microbial resistant germs survive and multiply.

Genotype, Chromosome and phenotype.

1. **Genome:** The genome is the entire set of DNAs found in a cell.
1. **Chromosome:** threadlike structures made up of protein and DNA that carry genomic information from cell to cell.
2. **Phenotype:** all observable characters that influenced by genotypes and environment

What Are Fimbriae? Give Location and Function?

Hair-like small appendages on G-neg cells

Composed of pilin, a protein

Can be at one pole or around the entire cell Used for attachment to surfaces of epithelial cell

- Fimbriae help bacteria attach to surfaces
- Fimbriae also help attach to epithelial cells e.g. Neisseria gonorrhoeae

Spontaneous Generation:

Aristotle: Simpler invertebrate could arise by spontaneous generation Francesco Redi challenged

Branches of Microbiology.

1. Bacteriology
2. Mycology
3. Basic and Applied microbiology
4. Medical microbiology
5. Epidemiology
6. Immunology
7. Food microbiology
8. Industrial Microbiology
9. Microbial Genetics & Mol. Biology

Five kingdom classification system.

In 1969 the five-kingdom classification was proposed by Robert Whittaker

- **Monera:** bacteria and archaea
- **Plantae:** plants
- **Animalia:** Animals
- **Protista**: they are unicellular eukaryotes.
- **Fungi:** yeast mold mushrooms, consist of chitin

Epidemiology and Its Type?

The study of where and when diseases occur and how they are transmitted in a population. There are three basic types of investigations relating to epidemiology

1. Descriptive Epidemiology
2. Analytical Epidemiology
3. Experimental Epidemiology

Probe and Its use in Biotechnology?

A probe is a single-stranded sequence of DNA or RNA used to search for its complementary sequence in a sample genome. And its uses:

1. DNA probes can be used for diagnosing various diseases.
2. Rapid detection and identification of infectious agents
3. Use in PCR, qPCR and hybridization techniques for detection
4. Ensure high specificity and sensitivity
5. Detection of nonviable organism

What Is Western Blotting?

Western Blotting: We can use this technique for the detection of antigen in the serum. Microbial problems can be separated on SDS-PAGE by electrophoresis and the presence of these proteins can be detected by enzyme-tagged antibodies specific to those proteins. A color band will be seen where the specific protein (antigen) 1s present on the gel. Please remember that proteins in the gel are first transferred to a paper strip before they could be detected by specific antibodies.

Factor determining bacterial composition:

Bacterial composition is influenced by environmental forces like pH, temperature, and availability. Host factors such as immune response and genetics also play a role. These factors shape microbial communities and their functions within a particular ecosystem or host environment.

How bacteria get N

1. Decomposing proteins
2. NH4+ ions
3. Nitrates
4. Gaseous N2
5. Nitrogen fixation

Components of lipoproteins:

Lipoprotein particles contain variable proportions of four major elements: cholesterol, triglycerides, phospholipids and specific proteins called apoproteins.

How antibodies are effective against microbes?

Antibodies help the body to fight microbes or the toxins (poisons) they produce. They do this by recognizing substances called antigens on the surface of the microbe, or in the chemicals they produce, which mark the microbe or toxin as being foreign. The antibodies then mark these antigens for destruction. Antibodies have a long, successful and yet bumpy history of effectiveness against viruses and bacteria.

Germ Theory of Disease:

1. Treatments for many diseases were discovered before Pasteur.
2. Causes of the disease were unknown.
3. Yeast fermentation by Pasteur was a clue
4. 1860, Joseph Lister, applied phenol as disinfectants in treating surgical wounds and hand sanitizing.

5. In 1865, Pasteur found that silkworm disease caused by a protozoan.

PCR & its purpose:

After 30 cycles, enough targets DNA accumulate which could be detected by electrophoresis. o PCR provides a quicker way of identification of microbes.

Ethylene oxide:

Ethylene Oxide: Many heat-sensitive items such as disposable plastic, petri dishes and syringes, heart-lung machine components, sutures, and catheters are sterilized by ethylene oxide. Ethylene oxide is a strong alkylating agent that kills by reacting with functional groups of DNA and proteins to block replication and enzymatic activity. It rapidly penetrates packing materials, even plastic wraps. EtO is explosive, supplied in a 10% to 20% concentration mixed with either CO2 or dichlorodifluoromethane. A clean object can be sterilized if treated for 5 to 8 hours at 38°C or 3 to 4 hours at 54°C when the relative humidity is maintained at 40 to 50% and the EtO concentration at 700 mg/L.

Chemical Methods for Microbial Control:

Chemicals are used to control microbial growth. Factors that influence the efficacy of a chemical disinfectant include the concentration of the disinfectant, presence of organic matter in the environment (matrix) where the disinfectant is being used, pH of the environment (disinfectants are more effective at acidic pH), and time of exposure to the disinfectant.

Three differences between prokaryotes and eukaryotes.

Prominent nucleus is found in eukaryotes while it is absent in prokaryotes. o Membrane bound organelles are found in eukaryotes and absent in prokaryotes. o No histones are found in prokaryotes while present in eukaryotes.

Examples prokaryote bacteria and archaea all organism found in domain Eukarya are eukaryotes.

Endospores:

- Specialized resting cells
- Highly durable, dehydrated forms of bacteria
- Formed inside a bacterium
- Survive extremes of temperature

Write the name of five layers of endospore o

1. **Exosporium:** A thin delicate outermost covering of the spore
2. **Spore coat:** 2nd layer underneath the Exosporium. It is thick and composed of several protein layers.
3. **Cortex:** It is the 3rd layer from outside in. It has peptidoglycan in it.
4. **Spore cell wall or core wall:** Surrounds the protoplast or spore core o **Spore core:** Contains nucleoid and ribosomes

Bacteria get Sulfur from:

1. Sulfate ions
2. Hydrogen sulfide
3. Sulfur containing amino acids

Functions of cell wall.

Cell wall: gives support and protection against plasmolysis.

Cell wall is the most important layer in bacteria. It is rigid layer just outside the plasma membrane. Most important structure in prokaryotes. It provides shape to the organism. It provides protection from osmotic lysis. It is involved in pathogenicity (ability of the organism to cause the disease). Several antibiotics act on it.

Antisepsis, sepsis, sanitization definition.

Antisepsis: Removing pathogens from living tissue.

Sepsis: Sepsis is the extreme response of body to an infection. Microbial contamination A toxic inflammatory condition arising from the spread of an organism from a focus of infections. Lowering microbial counts on eating utensils

Sanitization: removal of bacteria from surface

Site of entry of microbes in host?

Portal of Entry

1. Mucus membrane:
2. Tubal organs
3. Respiratory system and digestive system as largest portals
4. Skin: It is the largest organ of the body in terms of the surface area.
5. Parenteral route: By injury
6. Preferred route of entry: Streptococci inhaled vs ingested

Pili and Its Functions?

These are hair-like structure composed of pilin, usually one to ten in number. Longer than fimbriae.

Used for attachment to:

1. Host cells
2. Bacteria
3. Used for DNA transfer from one bacterium to another during Conjugation (Sex pili)
4. function in twitching Motility
5. Gliding Motility

Life cycle of fungi.

Filamentous fungi can reproduce asexually by fragmentation of their hyphae. This fragmentation takes various names according to the species of the fungus. Spores can also be produced sexually by these fungi; however, this happens only when food is scarce and environmental conditions are not favorable for further growth of filamentous fungi.

Asexual spores' type:

1. **Conidia:** Chain of spores at the end of a conidiophore Aspergillus
2. **Arthroconidia:** spores formed by fragmentation of septate hypha
3. **Ballistoconidia:** spores as budding of the parent cells
4. **Chlamydoconidium:** spores formed by rounding and enlargement within a hyphal segment

Conjugation, Transformation and Transduction.

1. **Transformation:** Naked DNA in solution is transferred from one bacterium to another during this process.
2. **Conjugation:** Genes transfer process can be mediated by plasmids. Plasmids are extra chromosomal circular DNA fragments that replicate independently of bacterial chromosome. Conjugation requires direct cell to cell contact and cells have to be opposite mating types.
3. **Transduction:** Transferring of a gene from a bacterium to another bacterium via a virus is called transduction. o When a bacteriophage infects a bacterium such as E. coli, 1t replicates inside E. coli and also produces a protein coat (which is called a capsid) in which viral DNA is packed before the virus is released.

Primary and Secondary immune response.

When a person is exposed to an antigen (a microbe for example), the antigen is presented to B cells which start making antibodies. Initially, there are no detectable antibodies in the serum. However, after 3-5 days, IgM antibodies are secreted and then 7-10 days after, the same B cell

starts making IgG. The level of IgG rises to a level and then it starts declining. This is a typical response of an antigen exposure and is called **a primary immune response**.

It is interesting to note that a second exposure with the same antigen results in a heightened response in terms of very high levels of antibodies that are achieved in a very short time (within 2-7 days). This is called a **secondary immune response.** It is also called a memory or anamnestic response because memory cells are involved that ensure a quick response to the second exposure by the same antigen

Vector contact and Vehicle transmission.

1.Contact Transmission:

- **Direct:** person to person, Physical contact, Influenza
- **Indirect:** A nonliving object is involved
- **Fomite:** any nonliving object (towel, bedding)
- **Droplet:** mucus droplets

2. Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include blood and other body fluids, drugs, and intravenous fluids.

3.Vectors: A vector is a vehicle (animate or inanimate) that carries a pathogen from one host to another. Arthropods are vectors for many diseases. There are two ways these arthropods can transmit organisms to susceptible hosts:

4= Mechanical Transmission: In this type of transmission, organisms stick to the body parts of insects and get transmitted passively from one location to another.

5= Biological Transmission: Microbes multiply in the insect and is transmitted actively by insect bites from one individual to another.

Causes and cure of Hypersensitivity type II.

This hypersensitivity involves antigenic cells and IgG or IgM antibodies that become reactive with these body cells. Whenever, IgM or IgG antibodies bind to their antigens, they always activate complement which then can punch holes in the cells lysing them altogether. Blood transfusion reaction is a good example of type II hypersensitivity.

Diagnosis: Cell deficiencies, Low number of RBCs due to hemolysis, Thrombocytopenia if platelets are involved.

Treatment: Avoid those drugs that cause it. Blood transfusion with correct blood type, Symptomatic treatment, treated according to the cause

Coenzyme with example.

Coenzymes are organic nonprotein molecules that bind with the protein molecule

(apoenzyme) to form the active enzyme (holoenzyme). Coenzymes such as NAD, NADP, FAD, FMN etc. are used as carriers of these electrons in oxidation reduction reactions

Hypersensitivity and Its Type?

It is a normal Immune response but in a damaging way. In other words, immune system is acting in a way for which it has made to. So, when an antigen gets into the body, immune system recognizes it as foreign and start making antibodies against the antigen.

Allergy is another name for hypersensitivity. Those antigens that act in a bad way are called allergens.

Types of Hypersensitivity:

Hypersensitivities can be grouped into 4 types based on the types of antibodies or cells involved.

1. **Type I (Anaphylactic**): Also called anaphylaxis or immediate hypersensitivity. This is IgE mediated. Happens in <30 mints.
2. **Type II (Cytotoxic**): IgG and IgM mediated as these antibodies attach to cells and destroy them by activating complement.

3. **Type III (Immune complex mediated)**: Immune complexes are formed by IgM or IgG antibodies and these complexes lodge in the capillary beds and cause damage by complement activation.
4. **Type IV (delayed type cell-mediated):** CD4+ helper cells along with macrophages are involved.

Adaptive Immunity or Difference Between Innate and Adaptive Immunity.

Immunity: Ability of the body to protect itself from harmful effects of diseases.

Two types:

Innate Immunity

Adaptive immunity

Innate Immunity:

Plants and animals have what is called innate immunity. Innate immunity is the first line of defense against pathogens. It involves several cell types, proteins, and even an organ. The organ involved is your skin. Yes, skin is part of the first line of defense. It protects you and prevents pathogens from getting inside your body.

Naturally Acquired Active Immunity: This type of adaptive immunity develops in a person who gets infected with a microbe but then recovers from that illness making antibodies. o **Naturally Acquired Passive Immunity:** A typical example of this type of immunity is transfer of antibodies from colostrum or milk of a mother to the offspring.

Adaptive immunity:

Adaptive immunity is an important part of the immune system. It is protection from an infectious disease agent that is mediated by B- and T- lymphocytes following exposure to specific antigen, and characterized by immunological memory.

Artificially Acquired Active Immunity: When a person receives a vaccine against a microbe and the body makes antibodies in response to that vaccine, it is called artificial acquired active immunity because body makes these antibodies actively by itself.

Artificially Acquired Passive Immunity: When premade antibodies are given to a person in order to protect a person from an illness or a snake bite, this constitutes an artificially acquired passive immunity type.

Role of fungi and its basic features.

Fungi (singular: fungus) are a kingdom of usually multicellular eukaryotic organisms that are heterotrophs (cannot make their own food) and have important roles in nutrient cycling in an ecosystem. Fungi reproduce both sexually and asexually, and they also have symbiotic associations with plants and bacteria. However, they are also responsible for some diseases in plants and animals. The study of fungi is known as mycology.

Function:

The cell wall of fungi is a dynamic structure that protects the cell from environmental stress and prevents from various effects that are caused due to changes in osmotic pressure.

Magnetosomes

- Inclusions of iron oxide
- Surrounded by invaginations of plasma membrane
- Present in G negative bacteria
- Act like a magnet. Bacteria can stick to iron containing rocks for nutrition. Decompose H2O2 which is toxic for cells.

Cause, Diagnose and treatment of Anaphylaxis?

Anaphylaxis is a severe, potentially life-threatening allergic reaction. It can occur within seconds or minutes of exposure to something you're allergic to, such as peanuts or bee stings

Causes: The patient also experiences difficulty in breathing. It is an emergency as it may kill the person. Anaphylaxis may be local or systemic. Shock from drug reactions, venoms, and common allergens causing asthma are some examples of type I hypersensitivity.

Diagnose: Diagnosed by demonstration of IgE by intradermal injections of allergens one by one. If specific IgE are present in the serum, a reaction will be visible on the skin locally in the form of a wheel of inflammation.

Treatment: Epinephrine & antihistaminic drugs provide a means of treatment.

Enlist four Phases of bacterial growth curve.

1. **The Lag Phase:** initial period of bacterial cells in new environment before starting growth.
2. **The Log Phase:** in this phase bacteria growth occur and cell number increase o **The Stationary Phase:** In the stationary phase, growth reaches a point in which number of dying cells equals the number of dividing cells. This is also called a period of equilibrium
3. **The Death Phase:** The number of death cells increase and number of new cells decreased in this phase is called death phase.

Taxonomic Hierarchy

Microbes are placed in groups based on similarities that they share with each other. All organisms can be grouped into a series of subdivisions that make up the taxonomic hierarchy. A bacterial species represents ―a monophyletic and genomically coherent cluster of individual organisms that show a high degree of overall similarity with respect to many independent characteristics, and is diagnosable by a discriminative phenotypic property‖ (definition taken from the internet).

A genus consists of various specie; however, these species differ from each other in certain ways, although these are related by descent with each other. Related genera make up a family. A group of similar families constitutes an order, and a group of similar orders makes up a class. Related classes, in turn, make up a phylum. All phyla that are related to each other make up a kingdom, and related kingdoms are grouped into a domain.

Domain ➢ Kingdom ▪ Phylum • Class Order ➢ Family ▪ Genus • Species

Photoautotrophs:

- ❖ Use light as a source of energy
- ❖ CO_2 as a source of Carbon
- ❖ Hydrogen atoms of water are used to reduce CO_2

Photoautotrophs: Oxygen is given off (Oxygenic)

- ❖ Photosynthetic bacteria
- ❖ Algae
- ❖ Green plants

Green bacteria: Non-oxygenic

- ❖ Do not use hydrogen from water
- ❖ Use hydrogen from H_2S or H_2 gas to reduce CO_2
- ❖ Need total anaerobic conditions

- ❖ Use light as a source of energy
- ❖ Use organic compounds as Carbon source
- ❖ Alcohol – Fatty acids – Carbohydrates etc.
- ❖ They **are anoxygenic.** – Rhodopseudomonas

Explain ELISA.

There are some interactions (Antigens and Antibodies) that are not directly evident. Interactions may have taken place, but there is no visible direct clue if the interaction has taken place or not. For such interactions, we use indirect methods.

Enzyme-Linked Immunosorbent Assay:

This test is done by coating the ELISA plastic plates either with the antigen or the antibodies. Suppose we have Brucella antigen coated ELISA plates that can detect antibodies from the serum of a Brucella infected animal. You take the serum and put into the wells of ELISA plate. You allow time for antigen and antibody to interact and then wash these wells with some buffer to remove unbound antibodies. These bound antibodies are not visible at this step. So, how do we see this interaction? Well, we use another antibody that has been made against the antibody bound to the antigen. This special antibody has been tagged with an enzyme. After allowing time for interaction of this 2ndantibody with the 1stantibody, 2naantibody is also given a washing to remove unbound antibody molecules. Again, we cannot say if interaction has taken place or not. Here comes an indirect way of detecting this interaction.

who discover small pox in which year?

Edward Jenner discovered small pox in 1796.

Factors that influence death rate of bacteria:

Six factors that affect microbial death are:

1. Osmotic pressure
2. Temperature
3. pH
4. Concentration of oxygen
5. Moisture content
6. Mechanical stress

Three basic types of bacteria:

The three basic types of bacteria are bacillus (rod shaped), coccus (spherical shaped), and spirillum (spiral shaped).

Which protein remain permanently in plasma?

Integral protein remains permanently in plasma.

Anti-viral drugs and their uses:

Antiviral Drugs

Nucleoside and nucleotide analogs

- Acyclovir
- Ganciclovir
- Ribavirin

Used mostly for herpesviruses

- Attachment and uncoating (Flu virus)
- Amantadine
- Zimantadine

Antifungal Drugs:

Amphotericin B

Ketoconazole

How infection occur in our body?

An infection occurs when germs enter the body, increase in number, and cause a reaction of the body. Three things that are necessary for an infection to occur: Source: Places where infectious agents (germs) live (e.g., sinks, surfaces, human skin) Susceptible person with a way for germs to enter the body.

Write functions of cell wall, flagella and ribosomes.

- **Cell wall:** gives support and protection against plasmolysis.
- **Flagella:** help bacteria in movement.
- **Ribosomes:** are organelles that assist in protein synthesis during translation.

Basic Features of Yeast:

- Non-filamentous fungi

- Unicellular
- Reproduce by budding (asexually) or sexually
- Some are dimorphic ¢ Larger than bacteria
- No flagella or cilia
- Facultative anaerobes
- Gram positive

Respiration is further divided into two types:

Aerobic Respiration: If in the electron transport chain, the final electron acceptor is oxygen, it is called an aerobic respiration.

Anaerobic Respiration: If in the electron transport chain, the final electron acceptor is any inorganic compound other than oxygen, it is called an anaerobic respiration. This final electron acceptor may be a nitrate ion, a sulfate ion, or a carbonate ion.

Disk Diffusion Method?

A standardized inoculum of the organisms is spread on agar plate.

Filter paper disks soaked with anti-bacterial

Plates are incubated at 37oC and results are recorded.

The size of the zone directly relates to the degree of sensitivity of the drug. This zone is compared to a standard table for recording the results of sensitivity test. Results are reported as sensitive or resistant or intermediate.

How Fungi Can Be Identified by Biochemical Testing?

Filamentous fungi are mainly identified by morphology, and not by biochemical testing. Yeast if grown on solid medium grows like bacteria. In other words, colonies of yeast look like bacterial colonies but are bigger in size.

Yeasts can grow as facultative anaerobes. Saccharomyces if given oxygen to grow, it converts carbohydrates aerobically and produces CO2 and water, however, if oxygen is denied, carbohydrates get converted fermentatively into ethanol and CO2.

How bacteria can be identified. Microbial identification:

Microbes especially bacteria (disease causing) can be identified by three methods:

A. Classical or Conventional Method B. Serological Methods C. Nucleic Acid based Methods Classical Methods

Classical Methods: of microbial identification involve differential staining of the sample before it its culturing, culturing the sample onto nutrient agar, blood agar and MacConkey's agar, purifying the culture (colonies expected to be involved in the disease) and detection of various enzymes that belong to various metabolic pathways. Classically, such methods used to take a long time to perform (3 days at least); however, rapid identification methods have become available now which use preformed media that can be used for testing the presence of metabolic enzymes.

Serological Methods: Serological Methods involve antibodies and antigen interactions. Antibodies are produced by B lymphocytes against any foreign antigens such as bacteria and their toxins that enter the body of animals or humans. Antibodies are very specific in their interaction. In other words, antibodies made against E. coli do not bind or interact with

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Staphylococci or vice versa. Although, there are many serological techniques that can be successfully used for identification of microbial infections, we will focus on only a few of them. The main advantage of using serological method is the speed and economy. In other words, serological methods are quick to do (take only about an hour) and can work directly on the sample (sample does not have to be cultured and purified as is needed in conventional methods of identification). Serological methods can be performed on cultured microbes as well which again speed up the diagnosis. Also remember that in all these serological tests, either the serum or the antigen should be known to us.

Write Names Of 5 Types of Bacteria with Respect to Temperature. s

Organisms are basically classified into three groups based on the temp requirements:

Psychrophiles: These are further divided into strict psychrophiles and Psychrotrophs:

Psychrotrophs: Cold loving: 15 oC

Psychrotrophs: Optimum temp 1s 20-30 oC Food spoilage bacteria that can spoil food during refrigeration.

Mesophiles: 25 — 40 oC: These are the ones that cause diseases in animals and humans. Moderate temp loving organisms.

Optimum: 37C as this is the body temperature of humans and animals.

Thermophiles: 50 — 60 oC o Heat loving o Important in organic compost piles. These are further divided into hyperthermophiles that grow optimally at 80 oC They live in hot springs. Every group of these organisms has a: o Minimum growth temp o Optimum growth temp. o Maximum growth temp.

Osmosis:

– Net movement of solvent from high conc of solvent to low conc – OR low conc of solute to higher conc of solute

Osmosis – Solvent will move b/c of semipermeable membrane

Osmotic Pressure – Pressure required to prevent movement of water into a solution containing some solute.

Explain Methods of Microbial Control in Detail.

These methods can be broadly classified into four groups: Physical, Mechanical, Chemical and Biological

Physical Methods of Microbial Control:

Heat: It is the cheapest sources of all and easily available to control microbial growth. Heat denatures enzymes: Methods that use heat can further be classified into dry and moist heat methods. Form of dry heat include:

Dry heat (hot-air oven),

Flaming (Platinum loop sterilization in the flame of Bunsen burner), Incineration (burning of ashes).

Pasteurization: This technique typically employs low heat for killing pathogenic and food spoiling bacteria in milk. The following three equivalent heat treatments can be given to achieve pasteurization of milk:

63°C for 30 min °

High-temperature short-time: 72°C for 15 sec

Ultra-high-temperature: 140°C for 4 sec

Filtration: It is also a physical method. It can be used to filter air (using high efficiency particulate air filter) or liquid medium using membrane filters.

Radiation: Ionizing and non-ionizing radiation Ionizing Radiation

Ionization radiations include gamma rays, X rays, or high-energy electron beams.

Non -ionizing Radiation: wavelengths longer than 1nm fall into non-ionizing radiation. The best example is UV light.

Chemical Methods for Microbial Control:

Chemicals are used to control microbial growth. Factors that influence the efficacy of a chemical disinfectant include the concentration of the disinfectant, presence of organic matter in the environment (matrix) where the disinfectant is being used, pH of the environment (disinfectants are more effective at acidic pH), and time of exposure to the disinfectant.

Medical microbiology? Medical Microbiology

Deals with diseases of humans and animals

Identifications of disease-causing agents

Control and elimination of microbes

Tracking down new pathogens

Who name scientifically to organism?

Linnaeus (in 1735) established the system of scientific nomenclature

Each organism has two names: the genus and specific epithet (descriptive word) – Escherichia coli.